

FL Simulation Results

File: test_A7

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 5
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 90.0% (0.90)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 90.0% (0.90) of data for 15 rounds with 5 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8755
Clean + DP Protection Accuracy: 0.8756
Poisoned Accuracy: 0.7834
Poisoned + DP Protection Accuracy: 0.7935
Drop (Clean - Poisoned): 0.0921
Drop (Clean - Init): 0.0827
Drop (Clean DP - Init): 0.0828
Drop (Poisoned - Init): -0.0094
Drop (Poisoned DP - Init): 0.0007
DP Protection Method: foolsgold
GPU Used: GPU 0

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8815

Clean Recall: 0.8755

Clean F1 Score: 0.8750

Clean DP Precision: 0.8815

Clean DP Recall: 0.8756

Clean DP F1 Score: 0.8751

Poisoned Precision: 0.7963

Poisoned Recall: 0.7834

Poisoned F1 Score: 0.7829

DP Protection Precision: 0.8070

DP Protection Recall: 0.7935

DP Protection F1 Score: 0.7926

Summary

FL Simulation Complete

Task: f811290f-f538-44ed-bbc9-a91c2f8a9b83

GPU: GPU 0

Init Accuracy: 0.7928

Clean Accuracy: 0.8755

Clean DP Accuracy: 0.8756

Poisoned Accuracy: 0.7834

Data Poison Protection Accuracy: 0.7935

Drop (Clean - Poisoned): 0.0921

Drop (Clean - Init): 0.0827

Drop (Clean DP - Init): 0.0828

Drop (Poisoned - Init): -0.0094

Drop (Poisoned_DP - Init): 0.0007

Data Poison Protection Method: foolsgold

--- Confusion Matrix Metrics (Weighted Avg) ---

Clean Precision: 0.8815

Clean Recall: 0.8755

Clean F1 Score: 0.8750

Clean DP Precision: 0.8815

Clean DP Recall: 0.8756

Clean DP F1 Score: 0.8751

Poisoned Precision: 0.7963

Poisoned Recall: 0.7834

Poisoned F1 Score: 0.7829

DP Protection Precision: 0.8070

DP Protection Recall: 0.7935

DP Protection F1 Score: 0.7926